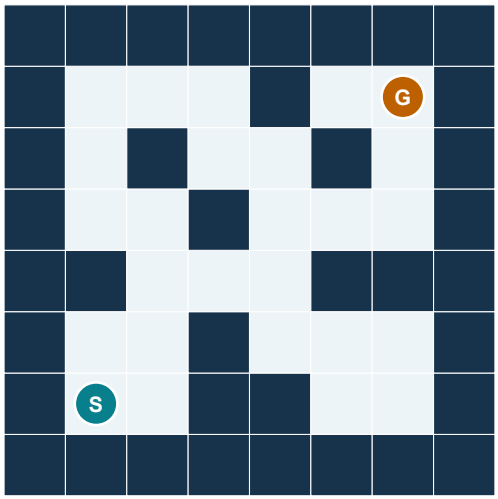
Standard PointMaze continuation

Separate continuation of the frozen November 5 study. Snapshot 2026-11-06T17:42:05.284432+00:00. DRAFT: active/unfinished work may remain.

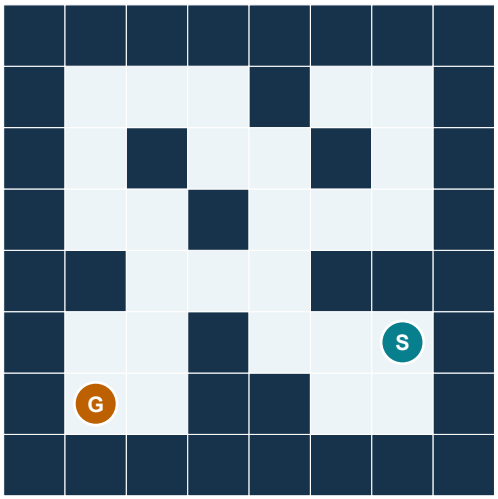
A point robot navigates around walls to a goal, with real next states and native rewards. Task 1 nominally starts at (0,0) and aims at (20,20); task 5 starts at (20,4) and aims at (0,0). Start coordinates have native reset noise; these are nominal centers, not measured trajectories.

Task 1: nominal start S / goal G



x = 4j - 4; y = 4i - 4 (up)

Task 5: nominal start S / goal G



x = 4j - 4; y = 4i - 4 (up)

Both standard tasks retain the native 1,000-decision limit and fixed single-task goals. PointMaze success uses distance at most 1 from the goal with the native single-task pre-step timing; rewards are -1 before success and 0 on success. This is separate from the earlier upstream-reset 128-step stress and from cube manipulation. The diagram has no heldout paths or outcomes.

The BC flow was learned from offline state-action data and takes x,y without an explicit goal input. Task-specific IQL-trained Q/V use real (s,a,r,s_next,done) tuples. Guidance uses frozen Q with a separately frozen BC actor; it is not an IQL advantage-weighted actor and Q is not ground truth.

Methods and coefficients

Twelve D-Flow families produce 22 fixed-lambda conditions; 3 QGF and 4 particle conditions make 29 guided rows. Eight controls remain explicit. Every task/model/condition chooses its own settings on validation only.

Family	Source updates	Jacobian / endpoint	Score
DF R full/full	regular	full / full	raw Q
DF R full/euler	regular	full / euler	raw Q
DF R identity/full	regular	identity / full	raw Q
DF R identity/euler	regular	identity / euler	raw Q
DF S full/full	stochastic	full / full	raw Q
DF S full/euler	stochastic	full / euler	raw Q
DF S identity/full	stochastic	identity / full	raw Q
DF S identity/euler	stochastic	identity / euler	raw Q
DF avgQ R full/full	regular	full / full	avg-Q
DF avgQ S full/full	stochastic	full / full	avg-Q
DF MH full/full	stochastic	full / full	raw Q + MH
DF avgQ MH full/full	stochastic	full / full	avg-Q + MH

D-Flow: z_next = z + eta [kappa g(z) - lambda z] + sqrt(2 eta T) eps. Regular variants omit movement noise; primary stochastic T=1. Lambda=0/1 is fixed per stratum, pulling toward origin when active. I is outer source iterations, not flow slices or physical steps.

Full-J/full guidance differentiates Q(F(z)); full-J/Euler differentiates the one-Euler proxy Q(E(z)), but both execute F(z). Identity variants replace the endpoint Jacobian with identity. QGF refreshes velocity, Euler endpoint and Q-gradient at every one of 8 pseudo-time slices; it has no eta/lambda/I source loop.

QGF: E_t = clip[a_t + (1-t) v(s,a_t,t)]; a_(t+dt) = a_t + dt [v(s,a_t,t) + kappa grad_a Q(s,E_t)], dt=1/8. Velocity, endpoint and action-Q gradient are evaluated fresh each slice; the final action is clipped. Original eight D-Flow variants use full/Euler guidance but all execute the complete clipped F8 after their outer source loop.

Averaged score: Qbar(s,a) = mean_k min_head Q_head(s,clip[a + sigma eps_k]). Its finite antithetic panel is fixed within an action call, including both MH sides. This changes the score; source movement noise changes the proposal/search.

Action-space Q averaging fixes an antithetic K=4/16 panel for an entire action call; sigma sets its action-space perturbation. Stochastic source noise, paper Jacobian smoothing and MH are distinct. MH proposes and accepts/rejects for a specified frozen-Q source target; it does not repair Q or guarantee finite-chain mixing. Numeric kappa is not comparable guidance strength across methods or critics.

IPG transports a particle group; CV adds its implemented control variate; RW retains residual importance weights; FK reweights/resamples without IPG transport. Final selection is uniform for IPG/CV and weighted for RW/FK. Particle noise, N and flow slices M are separate from D-Flow I.

Task 1: all guided conditions and controls

Requested condition	Seed 11	Seed 23	Seed 37
QGF	13/20	pending	pending
DF R full/full L0	14/20	pending	pending
DF R full/full L1	16/20	pending	pending
DF R full/euler L0	0/20	pending	pending
DF R full/euler L1	0/20	pending	pending
DF R identity/full L0	12/20	pending	pending
DF R identity/full L1	15/20	pending	pending
DF R identity/euler L0	13/20	pending	pending
DF R identity/euler L1	13/20	pending	pending
DF S full/full L0	12/20	pending	pending
DF S full/full L1	12/20	pending	pending
DF S full/euler L0	15/20	pending	pending
DF S full/euler L1	0/20	pending	pending
DF S identity/full L0	10/20	pending	pending
DF S identity/full L1	10/20	pending	pending
DF S identity/euler L0	12/20	pending	pending
DF S identity/euler L1	12/20	pending	pending
DF avgQ R full/full L0	13/20	pending	pending
DF avgQ R full/full L1	13/20	pending	pending
DF avgQ S full/full L0	12/20	pending	pending
DF avgQ S full/full L1	partial	pending	pending
DF MH full/full	pending	pending	pending
DF avgQ MH full/full	pending	pending	pending
QGF avg-Q	pending	pending	pending
QGF smooth-J	pending	pending	pending
IPG	pending	pending	pending
IPG-CV	pending	pending	pending
IPG-RW	pending	pending	pending
FK	pending	pending	pending
Canonical BC	0/20	pending	pending
Uniform BC N8_T8	0/20	pending	pending
Uniform BC N8_T32	0/20	pending	pending
Uniform BC N16_T8	0/20	pending	pending
Uniform BC N16_T32	0/20	pending	pending
QGF fixed 0.02	0/20	pending	pending
QGF fixed 0.04	0/20	pending	pending
Greedy 81-Q	10/20	pending	pending

Values are successes/20; completed zeros remain zeros. pending includes unfinished selection/testing; partial is an incomplete saved test batch. Blank CSV outcomes are never replaced by zero. A star is an exact config-ID display alias, not another physical trial. Three trained models share paired starts; there is no independent method-level sample from counting every row.

Task 5: all guided conditions and controls

Requested condition	Seed 11	Seed 23	Seed 37
QGF	pending	pending	pending
DF R full/full L0	pending	pending	pending
DF R full/full L1	pending	pending	pending
DF R full/euler L0	pending	pending	pending
DF R full/euler L1	pending	pending	pending
DF R identity/full L0	pending	pending	pending
DF R identity/full L1	pending	pending	pending
DF R identity/euler L0	pending	pending	pending
DF R identity/euler L1	pending	pending	pending
DF S full/full L0	pending	pending	pending
DF S full/full L1	pending	pending	pending
DF S full/euler L0	pending	pending	pending
DF S full/euler L1	pending	pending	pending
DF S identity/full L0	pending	pending	pending
DF S identity/full L1	pending	pending	pending
DF S identity/euler L0	pending	pending	pending
DF S identity/euler L1	pending	pending	pending
DF avgQ R full/full L0	pending	pending	pending
DF avgQ R full/full L1	pending	pending	pending
DF avgQ S full/full L0	pending	pending	pending
DF avgQ S full/full L1	pending	pending	pending
DF MH full/full	pending	pending	pending
DF avgQ MH full/full	pending	pending	pending
QGF avg-Q	pending	pending	pending
QGF smooth-J	pending	pending	pending
IPG	pending	pending	pending
IPG-CV	pending	pending	pending
IPG-RW	pending	pending	pending
FK	pending	pending	pending
Canonical BC	pending	pending	pending
Uniform BC N8_T8	pending	pending	pending
Uniform BC N8_T32	pending	pending	pending
Uniform BC N16_T8	pending	pending	pending
Uniform BC N16_T32	pending	pending	pending
QGF fixed 0.02	pending	pending	pending
QGF fixed 0.04	pending	pending	pending
Greedy 81-Q	pending	pending	pending

Values are successes/20; completed zeros remain zeros. pending includes unfinished selection/testing; partial is an incomplete saved test batch. Blank CSV outcomes are never replaced by zero. A star is an exact config-ID display alias, not another physical trial. Three trained models share paired starts; there is no independent method-level sample from counting every row.

Task 1: own validation-frozen settings

Requested guided condition	Seed 11	Seed 23	Seed 37
QGF	k80	k80	k40
DF R full/full L0	k5 e0.1 I256 L0	k20 e0.1 I256 L0	k5 e0.1 I256 L0
DF R full/full L1	k5 e0.03 I256 L1	k80 e0.01 I128 L1	k80 e0.03 I256 L1
DF R full/euler L0	k1 e0.1 I256 L0	BC fallback	BC fallback
DF R full/euler L1	k5 e0.1 I4 L1	k0 e0.01 I32 L1	BC fallback
DF R identity/full L0	k5 e0.1 I256 L0	k40 e0.1 I256 L0	k5 e0.1 I256 L0
DF R identity/full L1	k5 e0.1 I128 L1	k80 e0.03 I256 L1	k80 e0.03 I256 L1
DF R identity/euler L0	k20 e0.1 I256 L0	k20 e0.1 I256 L0	k5 e0.1 I256 L0
DF R identity/euler L1	k80 e0.03 I128 L1	k80 e0.03 I256 L1	k80 e0.03 I256 L1
DF S full/full L0	k80 e0.005 I256 L0	k40 e0.1 I128 L0	k80 e0.03 I128 L0
DF S full/full L1	k80 e0.005 I256 L1	k80 e0.01 I128 L1	k80 e0.01 I128 L1
DF S full/euler L0	k5 e0.1 I256 L0	k5 e0.1 I256 L0	k12 e0.1 I256 L0
DF S full/euler L1	k80 e0.1 I128 L1	k80 e0.1 I256 L1	k40 e0.1 I128 L1
DF S identity/full L0	k80 e0.01 I128 L0	k80 e0.03 I256 L0	k80 e0.1 I256 L0
DF S identity/full L1	k80 e0.01 I128 L1	k80 e0.03 I128 L1	k80 e0.03 I128 L1
DF S identity/euler L0	k80 e0.01 I128 L0	k80 e0.03 I256 L0	k80 e0.03 I128 L0
DF S identity/euler L1	k80 e0.01 I128 L1	k80 e0.03 I128 L1	k80 e0.03 I128 L1
DF avgQ R full/full L0	k40 e0.1 I256 L0 s0.2 K4	k80 e0.03 I256 L0 s0.2 K4	k20 e0.1 I256 L0 s0.2 K4
DF avgQ R full/full L1	k80 e0.03 I64 L1 s0.03 K4	k80 e0.01 I256 L1 s0.01 K16	k80 e0.01 I256 L1 s0.2 K4
DF avgQ S full/full L0	k80 e0.03 I256 L0 s0.2 K4	k80 e0.03 I128 L0 s0.1 K16	k80 e0.03 I128 L0 s0.01 K4
DF avgQ S full/full L1	k80 e0.03 I128 L1 s0.01 K16	k80 e0.01 I128 L1 s0.01 K16	k80 e0.01 I128 L1 s0.1 K4
DF MH full/full	k80 e0.1 I256 L1	k80 e0.1 I32 L1	k80 e0.1 I128 L1
DF avgQ MH full/full	k80 e0.1 I256 L1 s0.01 K16	k80 e0.1 I32 L1 s0.1 K16	k80 e0.03 I128 L1 s0.1 K16
QGF avg-Q	k80 s0.1 K16	k80 s0.01 K4	k80 s0.1 K16
QGF smooth-J	k40 s0.1 K4	k40 s0.2 K16	k80 s0.1 K16
IPG	k80 N8 M32 n0.005 r0.1	k40 N16 M32 n0.02 r0.01	k40 N16 M8 n0.02 r0.01
IPG-CV	k20 N16 M8 n0.02 r0.01	k40 N16 M8 n0.02 r0.01	k40 N16 M8 n0.02 r0.01
IPG-RW	k5 N16 M8 n0.005 r0.01	k40 N16 M32 n0.02 r0.1	k40 N16 M32 n0.02 r0.1
FK	k80 N16 M32 n0.02	k80 N16 M32 n0.02	k80 N16 M32 n0.02

k=kappa; e=eta; L=lambda; I=outer source iterations; s=sigma; K=Q averaging samples; N=particles; M=flow slices; n=particle noise; r=RKHS ridge. Unused quantities are omitted. BC fallback states the actual configuration while retaining the requested family.

CONTINUATION_SELECTED_SETTINGS.csv retains every exact JSON numeric value, primary and positive role, config ID and selection hash. CONTINUATION_TEST_RESULTS.csv includes all controls, frozen positive secondaries, exact-ID aliases, failed/partial cells and unused-parameter labels. No test result chooses a setting.

Task 5: own validation-frozen settings

Requested guided condition	Seed 11	Seed 23	Seed 37
QGF	k20	k40	k40
DF R full/full L0	k3 e0.1 I256 L0	not frozen	not frozen
DF R full/full L1	k5 e0.1 I128 L1	not frozen	not frozen
DF R full/euler L0	k1 e0.1 I256 L0	not frozen	not frozen
DF R full/euler L1	k80 e0.1 I16 L1	not frozen	not frozen
DF R identity/full L0	k1 e0.1 I256 L0	not frozen	not frozen
DF R identity/full L1	k5 e0.1 I128 L1	not frozen	not frozen
DF R identity/euler L0	k1 e0.1 I256 L0	not frozen	not frozen
DF R identity/euler L1	k40 e0.03 I64 L1	not frozen	not frozen
DF S full/full L0	k80 e0.01 I256 L0	not frozen	not frozen
DF S full/full L1	k80 e0.005 I128 L1	not frozen	not frozen
DF S full/euler L0	k80 e0.1 I128 L0	not frozen	not frozen
DF S full/euler L1	k80 e0.1 I256 L1	not frozen	not frozen
DF S identity/full L0	k80 e0.1 I256 L0	not frozen	not frozen
DF S identity/full L1	k5 e0.03 I128 L1	not frozen	not frozen
DF S identity/euler L0	k80 e0.1 I256 L0	not frozen	not frozen
DF S identity/euler L1	k5 e0.03 I128 L1	not frozen	not frozen
DF avgQ R full/full L0	k3 e0.1 I256 L0 s0.01 K16	not frozen	not frozen
DF avgQ R full/full L1	not frozen	not frozen	not frozen
DF avgQ S full/full L0	not frozen	not frozen	not frozen
DF avgQ S full/full L1	not frozen	not frozen	not frozen
DF MH full/full	not frozen	not frozen	not frozen
DF avgQ MH full/full	not frozen	not frozen	not frozen
QGF avg-Q	not frozen	not frozen	not frozen
QGF smooth-J	not frozen	not frozen	not frozen
IPG	not frozen	not frozen	not frozen
IPG-CV	not frozen	not frozen	not frozen
IPG-RW	not frozen	not frozen	not frozen
FK	not frozen	not frozen	not frozen

k=kappa; e=eta; L=lambda; I=outer source iterations; s=sigma; K=Q averaging samples; N=particles; M=flow slices; n=particle noise; r=RKHS ridge. Unused quantities are omitted. BC fallback states the actual configuration while retaining the requested family.

CONTINUATION_SELECTED_SETTINGS.csv retains every exact JSON numeric value, primary and positive role, config ID and selection hash. CONTINUATION_TEST_RESULTS.csv includes all controls, frozen positive secondaries, exact-ID aliases, failed/partial cells and unused-parameter labels. No test result chooses a setting.

Inherited work, new work and validation baselines

Execution origin	B8 batches	Completed rows
Inherited completed	615	4920
Mixed-origin continuation	1	8
Wholly new completed	264	2112

The inherited checkpoint already had 1,720 physics steps/calls across 8 active trajectories. Its completion is mixed-origin, not 8 fresh independent rollouts. Additional physics/calls in completed records: 1,260,343/1,260,343. Counters in current partial states remain unknown; 1 partial files were observed during this accounting capture.

Task / seed	Inherited rows	New whole rows	Mixed rows	Extra physics
1 / 11	1896	0	0	0
1 / 23	1896	0	0	0
1 / 37	1128	760	8	384758
5 / 11	0	1032	0	640975
5 / 23	0	160	0	135139
5 / 37	0	160	0	99471

Task / seed	BC /8	Positive QGF /8	Own kappa
1 / 11	0	8	80.0
1 / 23	0	8	80.0
1 / 37	0	8	40.0
5 / 11	0	8	20.0
5 / 23	0	3	40.0
5 / 37	0	8	40.0

These are 8-start validation controls, including failures. They are not heldout performance. Poor goal-reaching by goal-agnostic BC alone does not prove poor behavior-distribution fit. Source coverage, critic extrapolation, actor goal competence and numerical sensitivity remain distinct hypotheses.

Accounting capture: 2026-11-06T17:42:03.819625+00:00 to 2026-11-06T17:42:04.710033+00:00. It is staggered, not an atomic simulator snapshot. Additional-step totals exclude work still in partial checkpoints; inherited counts are not new experiments. Exact batch-origin and model tables accompany this report.

Actual heldout execution counts and measured batch time

Task / seed	Batches	Episodes	Physics	Active calls	Invalid events
1 / 11	28	560	370957	370957	0
1 / 23	0	0	0	0	0
1 / 37	0	0	0	0	0
5 / 11	0	0	0	0	0
5 / 23	0	0	0	0	0
5 / 37	0	0	0	0	0

Task / seed	Wall seconds	Sampler seconds	Timed batches W/S	Nonfinite returns
1 / 11	1418.374	1359.762	28/28	0
1 / 23	unmeasured	unmeasured	0/0	0
1 / 37	unmeasured	unmeasured	0/0	0
5 / 11	unmeasured	unmeasured	0/0	0
5 / 23	unmeasured	unmeasured	0/0	0
5 / 37	unmeasured	unmeasured	0/0	0

Task / seed	Complete display rows	Exact-ID aliases	Actual batches	Numeric groups?
1 / 11	28	0	28	28
1 / 23	0	0	0	0
1 / 37	0	0	0	0
5 / 11	0	0	0	0
5 / 23	0	0	0	0
5 / 37	0	0	0	0

Display rows include declared controls/primary conditions and any positive secondaries; aliases do not add physical time. Actual batches are unique config-ID/cohort files. Numeric-normalized groups are possible aliases, not proven identical policies. Repeated trajectory signatures are reported in the compute CSV and are not extra independent evidence.

Each cumulative batch wall/sampler timer repeats on all 20 episode rows: this report checks equality and counts it ONCE per physical batch. Wall starts after reset/restore; sampler timing includes inspect and host conversion, and compilation when inside it. Final serialization, model loading, training and offline prescreen are outside this measured test scope. Timers across simultaneous workers can overlap; their sum is not campaign elapsed time.

Only completed files contribute here. Partial computation remains unknown, not zero, and missing/nonfinite timer counts remain in CONTINUATION_TEST_COMPUTE.csv. Active calls and physics steps are measured counters; padded B20 work, inactive rows, AD, particles and flow/source iterations prevent treating them as measured Q-query counts or NFE. Invalid event totals retain zero where measured.

Provenance and limits

Task / model	Frozen conditions	Test batches / episodes	Task all-3 freeze
1 / 11	29	28 / 560	yes
1 / 23	29	0 / 0	yes
1 / 37	29	0 / 0	yes
5 / 11	18	0 / 0	pending
5 / 23	1	0 / 0	pending
5 / 37	1	0 / 0	pending

Scientific DESIGN 87063adcf6af771902e9bc6d4cfa4f0145ac14c4746ad94803dba26222556bab. New operational protocol 920ef8e10e27e73c8be096e23c1d3bffe52d09a0d84d8b15534228694d0ec5c9. Full checkpoint/config/source, selection and task-global-freeze hashes are in the snapshot and input/provenance CSVs. Every task freezes all 3 models x 29 selections before any of its 20-start tests. Unused original test cohorts were retained; the old campaign reports and archives remain frozen.

Local CPU only. The unchanged grid, per-method shortlists and inference budgets are unequal; validation uses simulator outcomes. I, M, particles, Q panels and AD incur different costs, so neither iteration counts nor active policy calls equal a measured Q/NFE count. Completed batch wall/sampler timers repeat on every episode row and must be counted once per physical configuration. No full training/prescreen/incomplete-work compute claim is made here.

Only 3 trained seeds and paired starts are available. Source noise, smoothing and MH do not identify true unseen-action Q or establish a universal winner. This descriptive report does not retune, rank-select, discard zero outcomes, or merge semantic/numeric aliases silently. Exact-ID display aliases are separate from actually repeated numeric-config executions.

Collection imports only stdlib and the separately reviewed stdlib accounting helper. PDF rendering uses the reporting runtime and Arial fonts; appearance requires root rendering and visual inspection of every page. Hash/schema checks do not rerun inference or prove causal explanations. Completed-file mtimes are artifact chronology, not an independent timestamp of first physics. Pending work stays pending.